Echoes in the Atmosphere: Reading Strangers' Poetic Traces in Place

HE LI, Guangzhou Academy of Fine Arts, China

This paper presents QUALIA, a physical–digital research probe for leaving and reading anonymous poetic traces in public space. Rather than treating memory as clearer capture or affect as a label to be detected, QUALIA explores how fleeting feelings may remain partially attached to the places where they were first experienced. In a small field probe on a campus bridge, participants first left poetic traces based on whispered feelings and situated impressions. Later visitors then returned to the same site, read these anonymous traces through a web-based map interface, and reflected on how the traces shaped their perception of the bridge. The study suggests that poetic traces may support a quiet form of asynchronous affective awareness by inviting later visitors to reconnect a stranger's feeling with the surrounding place.

Additional Key Words and Phrases: poetic traces, public space, place-based interaction, affective interaction, asynchronous awareness, research through design, human-AI interaction

1 Introduction

Public places are often remembered through small digital fragments. A lake is kept as a photograph, a bridge as a short video, a campus entrance as a location-tagged image in a phone album. These records make the scene easy to retrieve, but they also narrow experience into something mainly visual. What remains is often the view, the time, and the coordinate; what fades is the reason someone stopped, the atmosphere of the moment, and the feeling that made the place worth recording. Work on lifelogging and personal photography has similarly shown that digital records do not simply preserve experience, but shape what kinds of experience can later be selected, interpreted, and returned to [12, 15].

This issue becomes more subtle in public space. A bridge, a path, or a lakeside is shared by many people, yet the feelings attached to it are usually private and temporary. One person may pass through without noticing it, another may pause there because of the wind or light, and another may later remember it as a quiet emotional landmark. Following work on space and place, we treat place not as a fixed physical container, but as something made through situated use, memory, and interpretation [2, 7, 14]. From this view, a public site may hold many small affective encounters, even when these encounters are invisible to later visitors.

QUALIA explores how such encounters might be left behind and read again. It is a physical–digital research probe for creating anonymous poetic traces in public space. A person stops at a site, attends to the surroundings, and whispers a present feeling. The system combines this expression with situated cues, such as a blurred visual impression, time, and place context, to generate a short poetic trace. The trace is not meant to replace the moment with a clear photograph, nor to let AI speak completely for the user. It is closer to a partial residue of the encounter: something personal enough to carry a feeling, but open enough for someone else to approach.

This paper focuses on a small field probe conducted on a campus bridge. In the first stage, participants used QUALIA to leave poetic traces based on their immediate feelings in place. In the second stage, later visitors came to the same site, read these anonymous traces through a web-based map interface, and described how the traces affected their perception of the bridge. The study does not claim that poetic traces create strong social ties or stable collective memory. Instead, it asks a smaller question: how might a stranger's feeling become faintly perceptible when it is left as an ambiguous trace in the place where it was first felt?

Author's Contact Information: He Li, Guangzhou Academy of Fine Arts, Guangzhou, China, henryleecn0101@gmail.com.

2 Background and Design Motivation

The design of QUALIA began from a dissatisfaction with treating memory as a problem of fuller capture. Work on lifelogging has already questioned this assumption, arguing that the value of memory technologies lies not only in recording events, but also in supporting later selection and interpretation [12]. Research on physical mementos makes a related point: meaningful memory objects are often partial, ordinary, and context-bound, rather than complete records of what happened [11]. These works helped us move away from asking how public-space experiences could be recorded more accurately. We became more interested in how a record might remain incomplete in a useful way.

Before developing the probe, we conducted a small formative inquiry into everyday memory recording, including a questionnaire, semi-structured interviews, and a memory-recording probe. Participants often described photographs and short videos as useful anchors. They helped them recall where they were, who was present, and what the scene looked like. Yet when participants explained why a moment mattered, they usually added something outside the image: the weather, the sound of the place, a bodily feeling, a companion, or a mood that was difficult to see. The photograph was therefore rarely the whole memory. More often, it worked as a cue around which the felt experience had to be reconstructed.

This observation shaped the probe in three ways. First, recording should involve a pause, rather than reproduce the quick gesture of phone photography. This connects to slow technology, where interaction is designed not only for efficiency but also for reflection [6]. Second, the user should be able to express a feeling without choosing from fixed emotion labels. Here we follow critiques of affective computing that understand affect less as information to be detected, and more as something expressed and interpreted in interaction [1]. Third, the resulting record should not explain itself too completely, because later interpretation is part of the experience we wanted to study. QUALIA therefore uses whispering, blurred visual impressions, and AI-generated poetic language to make a different kind of record: not an objective account of a public place, but a situated trace that can be read, misread, and completed by someone who arrives later [5].

3 QUALIA as a Research Probe

QUALIA was built as a research probe to make this situation observable: a person leaves a feeling in a public place, and another person later encounters it there. The prototype was therefore not developed as a finished product, but as a way to stage a particular relation between body, place, expression, and later interpretation. This approach is informed by research through design and technology probes, where artifacts are used not only to solve predefined problems, but to open up, materialize, and study emerging design questions [8, 16]. In the field study, QUALIA combined a physical model, a web-based generation interface, and a map-based reading interface.

The physical model provided the bodily frame of the interaction. Participants held the object, looked toward the surrounding scene, turned the side knob, and spoke their feeling toward the device. These gestures were intentionally slower than taking a phone photograph. The model did not need to perform every function by itself; rather, it helped participants enter a different mode of recording, one that involved pausing, attending, and speaking from within the place. In this sense, the model worked less as a technical container and more as an embodied prompt for situated action [3].

The poetic trace was generated through a web prototype. A participant's whispered feeling was combined with situated cues, including time, place context, and a blurred visual impression of the scene. These materials were then used to generate a short poetic text. The aim was not to label the user's emotion or produce a complete literary poem, but to make a partial trace that still carried something of the original encounter. Each trace kept a degree of ambiguity, so that it could be read by a later visitor without fully determining what they should feel or understand [5].

Later visitors encountered these traces through a map-based interface. The map was not used mainly for navigation, but as a way to return to feelings attached to a place. Standing at the same site, a visitor could read anonymous traces left by previous participants, together with their blurred visual impressions. This hybrid setup allowed us to focus on the central question of the study: not whether QUALIA worked as a fully integrated device, but how a poetic trace might mediate between one person's moment of feeling and another person's later act of interpretation.

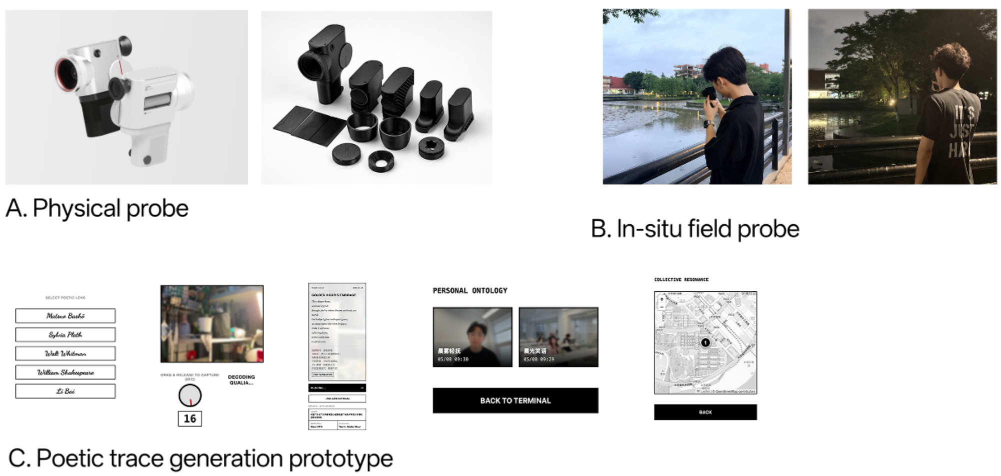

Fig. 1. QUALIA as a physical–digital research probe. (A) The physical probe frames the embodied act of pausing, looking, and whispering. (B) The in-situ field probe was conducted on a campus bridge where participants left and later read poetic traces. (C) The web-based prototype supports poetic trace generation, personal archive, and map-based reading.

4 Field Study: Reading Poetic Traces in Place

We conducted a small field probe on a pedestrian bridge near a university campus entrance. The site was chosen because it was both ordinary and affectively suggestive. For many students, it was simply a route into or across campus; at the same time, it offered water, wind, open views, vegetation, and ambient sound. This made it a suitable place to explore how a functional passage might become differently perceived when a stranger's feeling is encountered there.

Four students took part in the study. They were mutually unfamiliar and participated anonymously. The study was organized around two roles: leaving a trace and later reading a trace. In the first stage, participants stood on the bridge, attended to the surrounding scene, and used QUALIA to create poetic traces from their immediate feelings. The physical model framed the act of pausing and speaking, while the web prototype supported the actual generation of the trace.

In the second stage, later visitors came to the same bridge after a short delay. Before reading, they were asked to describe their ordinary impression of the site. They then opened the map-based interface and read anonymous poetic traces left by previous participants, together with blurred visual impressions of the scene. After reading, they described what they thought the traces expressed, how they connected them to the surrounding environment, and whether the traces changed how they perceived the bridge.

We collected field notes, generated traces, participants' spoken comments, and short post-use reflections. During analysis, we focused on how later visitors moved between the text and the place: which words they noticed, what parts of the environment they connected to the trace, and how their description of the site shifted after reading. Given the small scale and short delay, we treat this study as an exploratory probe rather than a validation of long-term collective memory. Its purpose was to see what becomes perceptible when a feeling is left in place and read there by someone else.

5 Findings

The field probe suggests that the traces were not read as isolated poems, but as something tied to the place where they appeared. Later readers moved back and forth between the text and the bridge: they looked at the water, noticed the wind and sound, and related certain words in the trace to what they could sense around them. In this way, the trace worked less like a message to be decoded and more like a cue that redirected attention to the site.

One effect was that the bridge became less anonymous. A participant who usually understood it as a passage described that, after reading another person's trace, the bridge felt less like "just a road" and more like a place that had carried someone's mood. This did not mean that the reader fully understood the original person's experience. Rather, the trace made the presence of an unknown other faintly perceptible. The site was no longer only a route through campus, but a place where someone else had once paused and felt something.

Ambiguity played an important role in this process. Readers did not always know exactly what the original trace leaver meant, and this uncertainty seemed to make interpretation easier rather than harder. The blurred image and poetic language left space for readers to connect the trace with their own memory of the bridge. One participant described the experience as being "like hearing someone else's echo in an everyday corner." The trace therefore did not transmit a fixed emotion from one person to another; it invited a situated form of completion.

Overall, the study points to a modest form of resonance. The poetic traces did not create direct conversation or lasting social ties. What they supported was weaker but still meaningful: a later visitor could stand in the same place, encounter a fragment of a stranger's feeling, and briefly re-sense the site through that fragment.

6 Discussion

The field probe suggests that poetic traces may support a form of public interaction that sits between private memory and direct communication. A trace is not a message addressed to a specific person, and it does not require a reply. At the same time, it is not purely private. Once attached to a place, it becomes available for someone else to encounter under similar surroundings. This makes the interaction quiet and indirect, but still social in a minimal sense. It resonates with prior work on awareness and familiar strangers, where social connection may arise not through direct conversation, but through peripheral cues, repeated presence, or the sensed existence of others in shared environments [4, 10].

For this kind of interaction, the role of AI is not to identify the user's true emotion. Nor is it simply to generate a better poem. Its value lies in making a fragile expression more shareable without making it too explicit. By turning a whispered feeling and situated cues into a short poetic trace, the system creates a medium that can travel from one moment of experience to another moment of interpretation. The trace keeps enough of the original atmosphere to be approached, but not so much that later readers are forced into a single meaning. In this sense, the poetic trace works as a small mediating object: stable enough to be encountered by different people, but open enough to support different interpretations [13].

This also points to the design value of ambiguity in public emotional expression. In public space, asking people to state their feelings directly may feel too exposed, while asking later visitors to understand those feelings accurately may be unrealistic. QUALIA instead works through partialness. The blurred image, anonymous authorship, and poetic language soften the emotional disclosure and allow readers to bring their own memory of the place into the interpretation. Resonance here is therefore not emotional transmission, but a situated act of reading with the place.

More broadly, this study suggests an alternative way to think about digital traces in public space. Instead of treating location-based media mainly as information, navigation, or social posting, QUALIA treats them as atmospheric residues: small fragments that can make the affective life of a place faintly perceptible. This connects to experience-centered approaches in HCI, where the value of technology is not limited to task completion, but also lies in how it shapes felt, situated, and interpretive experience [9]. Future designs might explore how such traces accumulate over longer periods, how people choose what to reveal or hide, and how public interfaces can support affective awareness without turning private feelings into public exposure.

7 Limitations and Future Work

This study is limited in scale. The field probe involved only four participants, one campus bridge, and a short delay between leaving and reading traces. The findings therefore should not be read as evidence that QUALIA can produce long-term collective memory or stable public connection. They show only an early glimpse of how anonymous poetic traces may be interpreted when they are read in the place where they were created.

The prototype was also a hybrid research setup rather than a fully integrated system. The physical model supported the bodily act of pausing and speaking, while the poetic generation and map-based reading were implemented through web interfaces. This separation allowed us to observe the interaction more clearly, but it also means that the study did not yet test a seamless end-to-end experience in everyday use.

Future work will involve longer deployments across multiple public sites, with more participants and longer intervals between leaving and reading traces. This would allow us to examine how traces accumulate over time, whether repeated encounters change people's sense of a place, and how different kinds of public spaces shape interpretation. We also plan to develop a more integrated prototype that supports situated capture, poetic generation, anonymous publishing, and later revisiting in one continuous interaction.

Finally, future studies need to address the ethics of public affective traces. Even when traces are anonymous and ambiguous, they may still carry sensitive feelings. Further work should explore how people choose what to reveal, how traces can fade or be removed, and how public interfaces can support emotional resonance without turning private experience into unwanted exposure.

8 Conclusion

This paper presented QUALIA, a physical–digital research probe for leaving and reading anonymous poetic traces in public space. Rather than treating memory as clearer capture or affect as a label to be detected, QUALIA explores how a fleeting feeling might remain partially attached to the place where it was first experienced.

Through a small field probe on a campus bridge, we examined how later visitors encountered poetic traces left by strangers at the same site. The study suggests that such traces can make the presence of an unknown other faintly perceptible, not by transmitting emotion directly, but by inviting readers to connect the trace with the surrounding environment and their own memory of the place.

QUALIA therefore points to a quieter form of public interaction: one based on pause, ambiguity, and delayed interpretation. A poetic trace does not need to become a message, a post, or a permanent monument. It can remain small and incomplete, yet still allow a later visitor to sense that a place has been felt before.

Generative AI Use Statement

During the preparation of this manuscript, the author used generative AI tools, including ChatGPT, to improve clarity, organization, and English expression. The author reviewed, revised, and verified all content, and takes full responsibility for the final manuscript.

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